

Revisiting the Technology Shock–Hours Debate

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Abstract

This report replicates the estimation of the effects of technology shocks using a structural VAR (SVAR) as in [Galí \(1999\)](#) and [Christiano, Eichenbaum, and Vigfusson \(2003\)](#), using updated data through 2025Q3. We find that standard SVAR methods reproduce the key result of the debate: the estimated effects of a technology shock depend on how per-capita hours are modeled. We then implement local projections to estimate impulse responses and find that the results are less sensitive to the specification of hours. In particular, technology shocks are associated with a positive short-run response of hours worked.

1 Introduction

This report contributes to the technology shock–hours worked debate initiated by [Galí \(1999\)](#) and reassessed by [Christiano, Eichenbaum, and Vigfusson \(2003\)](#). The debate concerns the response of hours worked to a permanent increase in labor productivity, i.e. a technology shock. Both contributions rely on the same long-run identifying idea: technology shocks are the only disturbances that can affect the long-run level of (log) labor productivity. In particular, while Real Business Cycle (RBC) models predict a pro-cyclical response of hours, [Galí \(1999\)](#) suggests that, in the presence of nominal rigidities, technology shocks may lead to a short-run decline in employment.

This report uses quarterly U.S. data from FRED on real output in the nonfarm business sector (OUTNFB), total hours (HOANBS), and the civilian noninstitutional population aged 16 and over (CNP16OV). From these series, we construct (log) labor productivity and per-capita hours.

The empirical analysis compares two specifications that mirror the core difference between the papers, namely how to treat h_t (as $I(1)$, i.e. difference-stationary, or as stationary in levels). For each specification, we estimate a VAR(4), compute Wold impulse responses, and identify the technology shock by imposing the long-run restriction that only the technology shock has a non-zero long-run effect on y_t (as in [Blanchard and Quah \(1989\)](#)). For each specification, then we report impulse response functions for y_t and h_t together with 68% bootstrap confidence bands, and compute the forecast error variance shares attributable to the identified technology shock. To facilitate comparison, we also implement local projections (LP), [Jordà \(2005\)](#) with lagged controls, compare VAR-based and LP-based impulse responses, assess sensitivity to longer lag lengths (12 and 16), and examine whether the responses differ between positive and negative technology shocks. Overall, this study provides a broad and robust assessment of the impact of technological innovations on the U.S. labor market.

2 Data

The dataset includes three U.S. macroeconomic series from FRED (Federal Reserve Bank of St. Louis):

- Real output in the nonfarm business sector, **OUTNFB** (quarterly, seasonally adjusted index; 2017=100).
- Total hours, **HOANBS** (quarterly, seasonally adjusted index; 2017=100).
- Working-age population, **CNP160V** (monthly, not seasonally adjusted; thousands of persons).

The target sample is 1948Q1–2025Q3 (quarter-start dates from 1948-01-01 to 2025-07-01). Because **CNP160V** is monthly, it is aggregated to quarterly frequency by averaging the three monthly observations within each quarter and aligning the result to the quarter-start date.

We then define

$$y_t = \log\left(\frac{\text{OUTNFB}_t}{\text{HOANBS}_t}\right) \quad (\text{log labor productivity})$$

$$h_t = \log\left(\frac{\text{HOANBS}_t}{\text{CNP160V}_t}\right) \quad (\text{log per-capita hours}).$$

From these series, we define the two specifications used in the literature.

- The Galí specification: $[\Delta y_t, \Delta h_t]'$
- the CEV specification uses $[\Delta y_t, h_t]'$

The series are plotted in Figure 1.

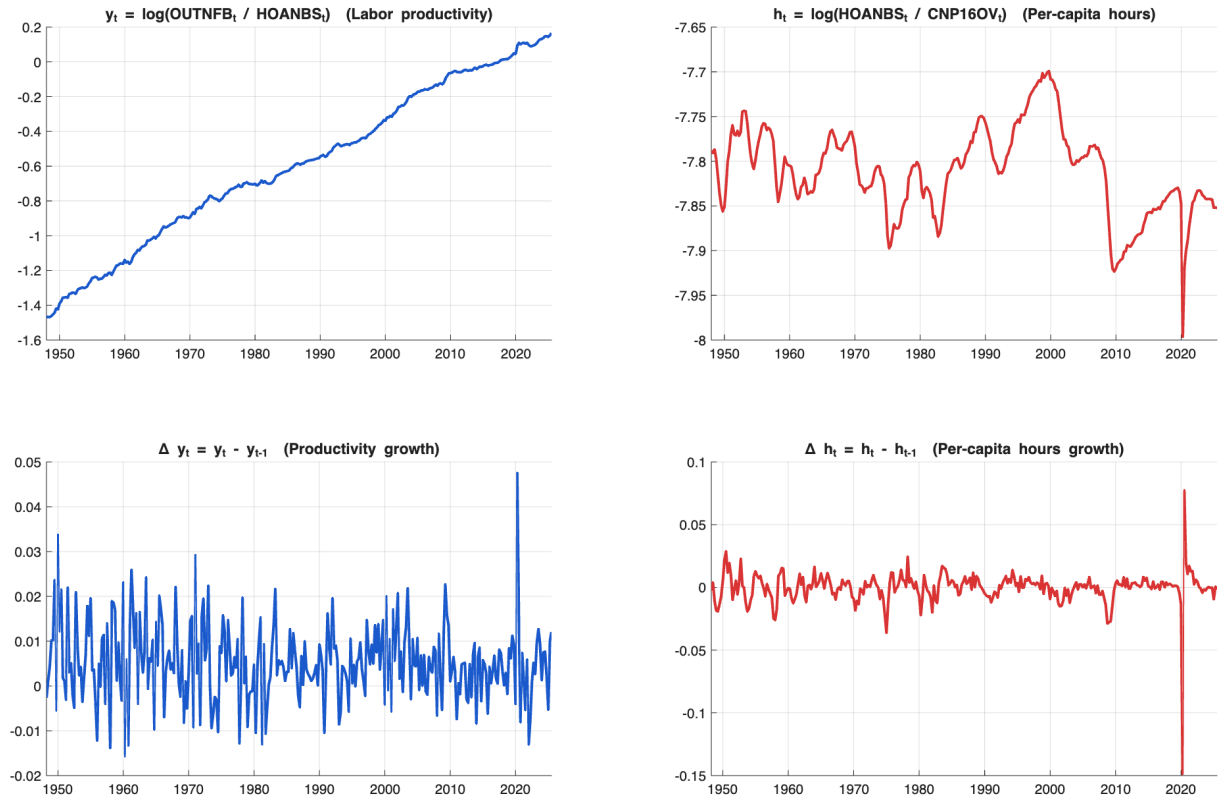


Figure 1: Labor productivity and per-capita hours (levels and differences).

Labor productivity y_t is clearly non-stationary, displaying an upward stochastic trend. A

preliminary inspection of h_t highlights a key modeling issue for the technology-shock-hours debate. Visually, h_t appears to fluctuate around a relatively stable long-run level, which is suggestive of a stationary ($I(0)$) or strongly mean-reverting process. At the same time, Δh_t also appears stationary, and the literature emphasizes that standard time-series diagnostics often have limited power to distinguish a highly persistent $I(0)$ process from a unit-root ($I(1)$) process. This motivates estimating both specifications.

3 Methodology

We replicate the analysis using a structural VAR. In particular, to obtain shocks that are economically interpretable, we identify the technology shock by imposing long-run restrictions, as in [Blanchard and Quah \(1989\)](#). We then implement local projections (LP) to estimate impulse response functions and compare SVAR- and LP-based results.

We now provide a brief recap of identification starting from a VAR(p) specification and the associated Wold impulse responses. Consider the reduced-form VAR(p)

$$Y_t = c + \sum_{j=1}^p A_j Y_{t-j} + u_t, \quad \mathbb{E}[u_t] = 0, \quad \mathbb{E}[u_t u_t'] = \Sigma_u,$$

where Y_t is either $[\Delta y_t, \Delta h_t]'$ or $[\Delta y_t, h_t]'$.

Under stability, the VAR admits the Wold moving-average representation

$$Y_t = \mu + \sum_{j=0}^{\infty} C_j u_{t-j}, \quad C_0 = I,$$

with long-run multiplier

$$C(1) \equiv \sum_{j=0}^{\infty} C_j.$$

The matrices $\{C_j\}_{j=0}^H$ provide the Wold (reduced-form) impulse responses.

Then, to implement the long-run identification of the technology shock as in Blanchard-Quah and identify structural shocks, we assume that the reduced-form innovations u_t are linear combinations of orthonormal structural shocks ε_t . Specifically, we write

$$u_t = B_0 \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I,$$

so that $\Sigma_u = B_0 B_0'$. Structural impulse responses are then

$$\Theta_j = C_j B_0, \quad j = 0, 1, \dots, H,$$

and the long-run impact matrix is $\Theta(1) = C(1)B_0$.

To identify the technology shock, we assume that non-technology shocks (e.g. demand or monetary shocks) have no permanent effect on the level of labor productivity y_t . In terms of long-run impacts, this implies that the long-run impact matrix is lower triangular. Let the structural shocks be ordered as

$$\varepsilon_t \equiv \begin{bmatrix} \varepsilon_t^{\text{tech}} \\ \varepsilon_t^{\text{non-tech}} \end{bmatrix}, \quad \mathbb{E}[\varepsilon_t \varepsilon_t'] = I.$$

Then

$$\Theta(1) = \begin{pmatrix} \theta_{y,\text{tech}}(1) & \mathbf{0} \\ \theta_{h,\text{tech}}(1) & \theta_{h,\text{non-tech}}(1) \end{pmatrix}.$$

The zero restriction $\theta_{y,\text{non-tech}}(1) = \mathbf{0}$ ensures that only the technology shock has a non-zero long-run effect on productivity, while the long-run effect on hours depends on how hours are modeled (in differences or in levels).

To impose this restriction operationally, we compute

$$M \equiv C(1)\Sigma_u C(1)',$$

take its Cholesky factor K such that $KK' = M$, and set

$$B_0 = C(1)^{-1}K.$$

This implies $\Theta(1) = C(1)B_0 = K$, which is lower triangular by construction and therefore satisfies the long-run recursive restriction.

To avoid numerical issues in computing VAR-based standard errors due to collinearity, we construct confidence bands for the VAR impulse responses via bootstrap methods in fact uncertainty for VAR-based IRFs is quantified via a residual bootstrap using a standard procedure.

We quantify the importance of the identified technology shock by computing the forecast error variance decomposition (FEVD). This metric measures the fraction of the h -step-ahead forecast error variance of labor productivity and hours worked that is attributable to the identified technology shock.

The contribution of the technology shock to the forecast error variance of variable i at horizon h is

$$\omega_{i,\text{tech}}(h) = \frac{\sum_{j=0}^{h-1} (e_i' \Theta_j e_{\text{tech}})^2}{\sum_{j=0}^{h-1} e_i' \Theta_j \Theta_j' e_i},$$

where e_i selects variable i and e_{tech} selects the technology shock.

3.1 Local projections (LP)

Impulse responses can also be estimated using local projections (Jordà, 2005). For a shock (or shock proxy) s_t , the IRF of variable y at horizon h is

$$R_{s \rightarrow y}(h) \equiv \frac{\partial \mathbb{E}[y_{t+h}]}{\partial s_t}.$$

To impose the same long-run identifying idea in the LP framework, we follow a two-step procedure in the spirit of Blanchard and Quah (1989). First, using a large horizon H_{LR} , we estimate

$$y_{t+H_{\text{LR}}} - y_{t-1} = \mu_{H_{\text{LR}}} + g'_{H_{\text{LR}}} w_t + \sum_{j=1}^p \phi_j' w_{t-j} + \eta_{t+H_{\text{LR}}}, (\text{step1})$$

and construct the technology-shock proxy $\hat{s}_t \equiv \hat{g}'_{H_{\text{LR}}} w_t$.

Here $w_t = [\Delta y_t, \Delta h_t]'$ in specification 1 and $w_t = [\Delta y_t, h_t]'$ in specification 2.

Second, for each variable $i \in \{y, h\}$ and each horizon $h = 0, 1, \dots, H$

with $H < H_{LR}$, we estimate

$$z_{i,t+h} - z_{i,t-1} = \alpha_{i,h} + \beta_{i,h} \hat{s}_t + \sum_{j=1}^p \gamma'_{i,h,j} w_{t-j} + \nu_{i,t+h}, \text{ (step2)}$$

where $z_{y,t} \equiv y_t$ and $z_{h,t} \equiv h_t$. Then $\{\beta_{i,h}\}_{h=0}^H$ traces the response of $z_{i,t}$ to the identified long-run technology shock.

3.2 Sign-dependent (nonlinear) local projections

To allow for different responses to positive and negative shocks, we estimate a sign-dependent LP specification based on the positive part of the identified shock. After computing the shock as in step 1 and normalizing it, we modify step 2 as follows. For each horizon $h = 0, 1, \dots, H$,

$$z_{t+h} - z_{t-1} = \alpha_h + \beta_h s_t + \beta_h^+ s_t^+ + \gamma_h' x_t + v_{t+h},$$

where

$$s_t^+ \equiv \max(s_t, 0).$$

This specification nests the linear case when $\beta_h^+ = 0$ and allows the impulse response to depend on the sign of the shock.

Given estimates $\hat{\beta}_h$ and $\hat{\beta}_h^+$, the impulse response of z at horizon h to a shock of size δ is computed as

$$R_{s \rightarrow z}(\delta, h) = \hat{\beta}_h \delta + \hat{\beta}_h^+ \mathbb{E}[(s_t + \delta)^+ - s_t^+], \quad (x)^+ \equiv \max(x, 0),$$

where the expectation is implemented by the sample average

$$\hat{\mathbb{E}}[(s_t + \delta)^+ - s_t^+] = \frac{1}{T} \sum_{t=1}^T \left(\max(s_t + \delta, 0) - \max(s_t, 0) \right).$$

Since, the identified shock is normalized, then $\delta = 1$ corresponds to a one-standard-deviation innovation. Confidence intervals for LP coefficients are computed using HAC standard errors.

4 Empirical Results

4.1 Galí Specification: $w_t = [\Delta y_t, \Delta h_t]'$

We first comment briefly on the results obtained under the Galí specification. After estimating the VAR and computing impulse responses, we cumulate the responses so that they can be read as effects on the levels. Figure 2 reports the cumulated Wold impulse responses from the VAR(4); however, these responses do not have a structural economic interpretation.

Figure 3 reports impulse responses to the identified technology shock obtained by imposing the long-run restriction discussed above: the non-technology disturbance has zero long-run effect on (log) labor productivity. As shown in the figure, a positive technology shock generates a permanent increase in labor productivity: the response of y_t is positive on impact and converges to a strictly positive long-run level. The response of per-capita hours is negative: h_t falls on impact and remains below zero at long horizons, although the 68% bands include zero at several horizons,

indicating non-negligible estimation uncertainty about the magnitude of the hours response in this specification. These results are consistent with Galí (1999) under this specification.

However, when we estimate the same effects using the LP specification, the response of productivity remains broadly similar, while the response of hours turns positive on impact, in line with Christiano, Eichenbaum, and Vigfusson (2003). This contrast is informative because local projections are less sensitive than VAR-based IRFs to dynamic misspecification and to the exact treatment of persistence in the system. In this sense, the LP evidence provides additional support for the CEV view of a positive short-run hours response to a technology shock. The LP results are plotted in Figure 4. Figure 5 overlays the SVAR- and LP-based impulse responses to facilitate comparison.

As for the forecast error variance decomposition, under this specification the identified technology shock explains about 62.29% of the variance of Δy_t and 43.08% of the variance of Δh_t . Table 1 reports the FEVD at different horizons, showing that the share of variance explained is larger at shorter horizons.

4.2 CEV Specification: $w_t = [\Delta y_t, h_t]'$

We repeat the same identification exercise using the alternative specification in which hours enter in levels. We find that while the reduced-form impulse responses are broadly similar (Figure 6), the responses change once the technology shock is identified (Figure 7). In particular, a positive technology shock raises hours on impact: h_t increases sharply in the first quarters and then gradually decays toward zero at long horizons. At the same time, productivity growth Δy_t remains positive and implies a positive long-run effect on the level of productivity.

This pattern is consistent with the LP results reported in Figure 8. In particular, the point estimate for h_t remains positive throughout the displayed horizons, although it also declines over time, suggesting a more persistent positive hours response under this specification. Overall, the LP evidence supports a positive short-run hours response, while longer-horizon effects are less precisely estimated, as reflected by the widening HAC bands. These findings are consistent with the CEV specification and with results reported in the literature when identification relies on external instruments for technology shocks. In Figure 9 we also overlays the SVAR- and LP-based impulse responses to facilitate comparison.

As for the forecast error variance decomposition, under this specification the identified technology shock explains about 85.92% of the variance of Δy_t and 34.63% of the variance of h_t . Table 2 reports the FEVD at different horizons. The share explained for Δy_t is larger at shorter horizons, while for h_t the share explained increases with the forecast horizon.

4.3 Non-Linearity and Robustness

Figures 10 report the sign-dependent LP IRFs for the two specifications, which allow the responses to differ after positive and negative technology shocks (with 68% HAC bands). The responses are strongly asymmetric. For productivity, a positive shock produces a persistent increase in y_t , while a negative shock produces a persistent decline. The two paths are not mirror images, indicating that the effects are not simply symmetric up to sign.

Hours display an even sharper asymmetry. After a positive technology shock, h_t increases on impact (hump-shaped in the short run) and then gradually declines toward zero. After a negative shock, h_t falls sharply and remains below zero for many horizons, with only partial mean reversion.

The qualitative patterns are similar across the two specifications, as expected given that local projections are less sensitive than VARs to the exact treatment of persistence in the system.

Overall, the nonlinear LP evidence suggests that the propagation of technology shocks depends on their sign, with negative shocks associated with larger and more persistent contractions in hours than the expansion generated by positive shocks.

Regarding the inclusion of additional lagged controls, Figures 11 and 12 show that the coefficients are stable for both y_t and h_t . This suggests that the output and hours dynamics are largely governed by the identified shock rather than by the particular lag length used to partial out predictable components. In particular, the results indicate that the technology shock has a positive effect on both h_t and y_t , especially in the long run. In the short run, however, the results for y_t are more ambiguous: in the second specification, when using 12 or 16 lags, the short-run effects are not statistically significant.

5 Conclusion

This report revisits the technology shock–hours worked debate using quarterly U.S. data through 2025Q3. As expected in a bivariate structural VAR identified by the Blanchard–Quah long-run restriction, the estimated response of hours depends on how per-capita hours are modeled. Consistent with Galí (1999), when hours enter in first differences, a positive technology shock is associated with a decline in hours. In contrast, when hours enter in levels, we obtain results consistent with CEV (Christiano, Eichenbaum, and Vigfusson, 2003), with hours increasing on impact and then gradually returning toward zero.

We also estimate impulse responses using local projections based on a long-run technology-shock proxy constructed in a two-step procedure. The LP-based responses are less sensitive to whether hours are treated in differences or in levels. Across both specifications, technology shocks are associated with positive medium-run effects on labor productivity and a positive short-run response of hours, although longer-horizon estimates become less precise. Overall, the LP evidence points to a persistent positive effect of technology shocks, in line with more recent studies that identify technology shocks using external instruments.

Finally, sign-dependent local projections indicate substantial asymmetries. Negative technology shocks generate persistent contractions in productivity, while positive technology shocks are associated with a persistent expansion in hours. The response of hours to negative shocks is more mean-reverting, gradually returning toward zero at longer horizons.

6 Appendix

6.1 Figures

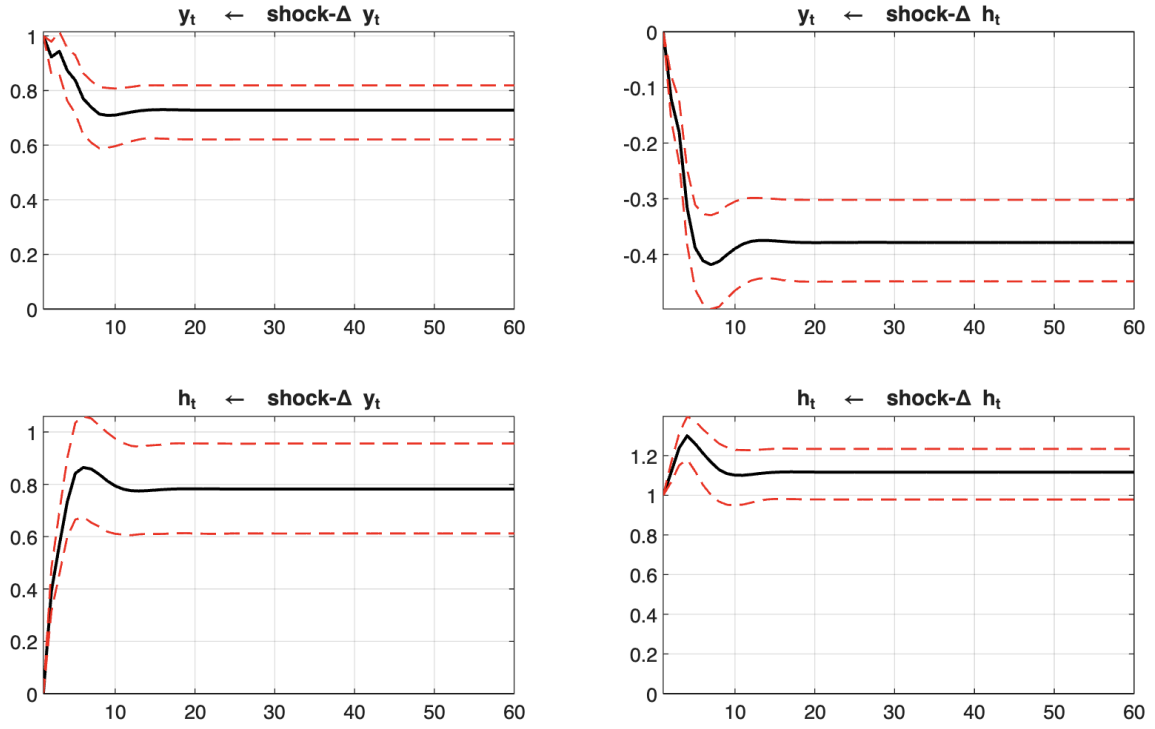


Figure 2: Cumulated Wold Impulse Responses from the VAR(4) with $w_t = [\Delta y_t, \Delta h_t]'$
Confidence Interval: Bootstrap 68%

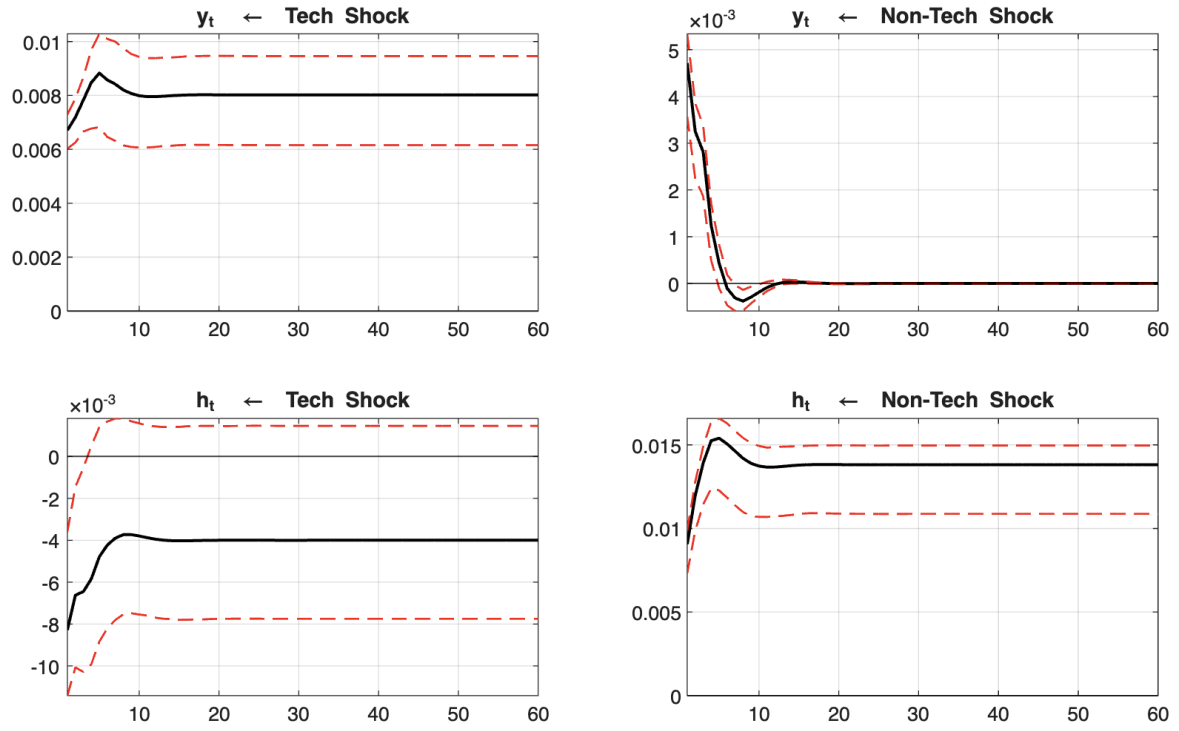


Figure 3: Cumulate Impulse responses to the Identified Technology Shock with $w_t = [\Delta y_t, \Delta h_t]'$
Confidence Interval: Bootstrap 68%

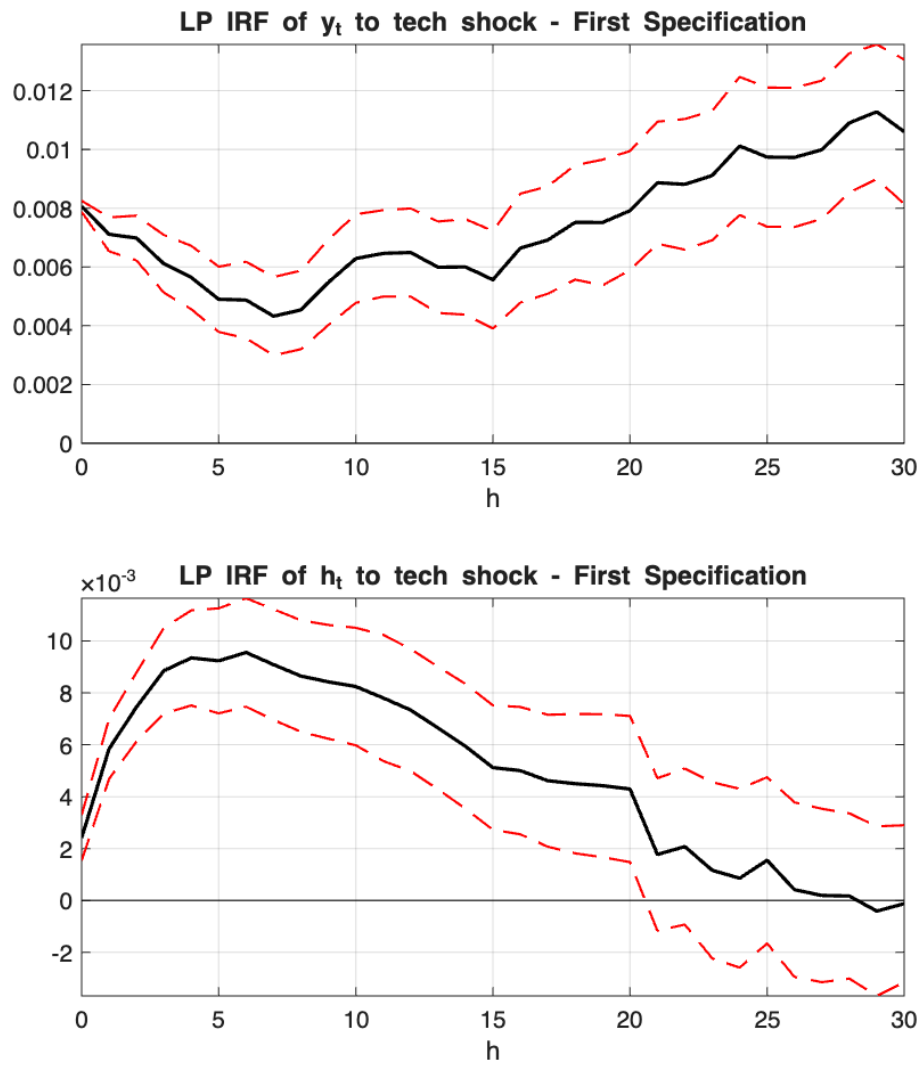


Figure 4: Linear Projection, for the first specification. Confidence Interval: HAC 68%

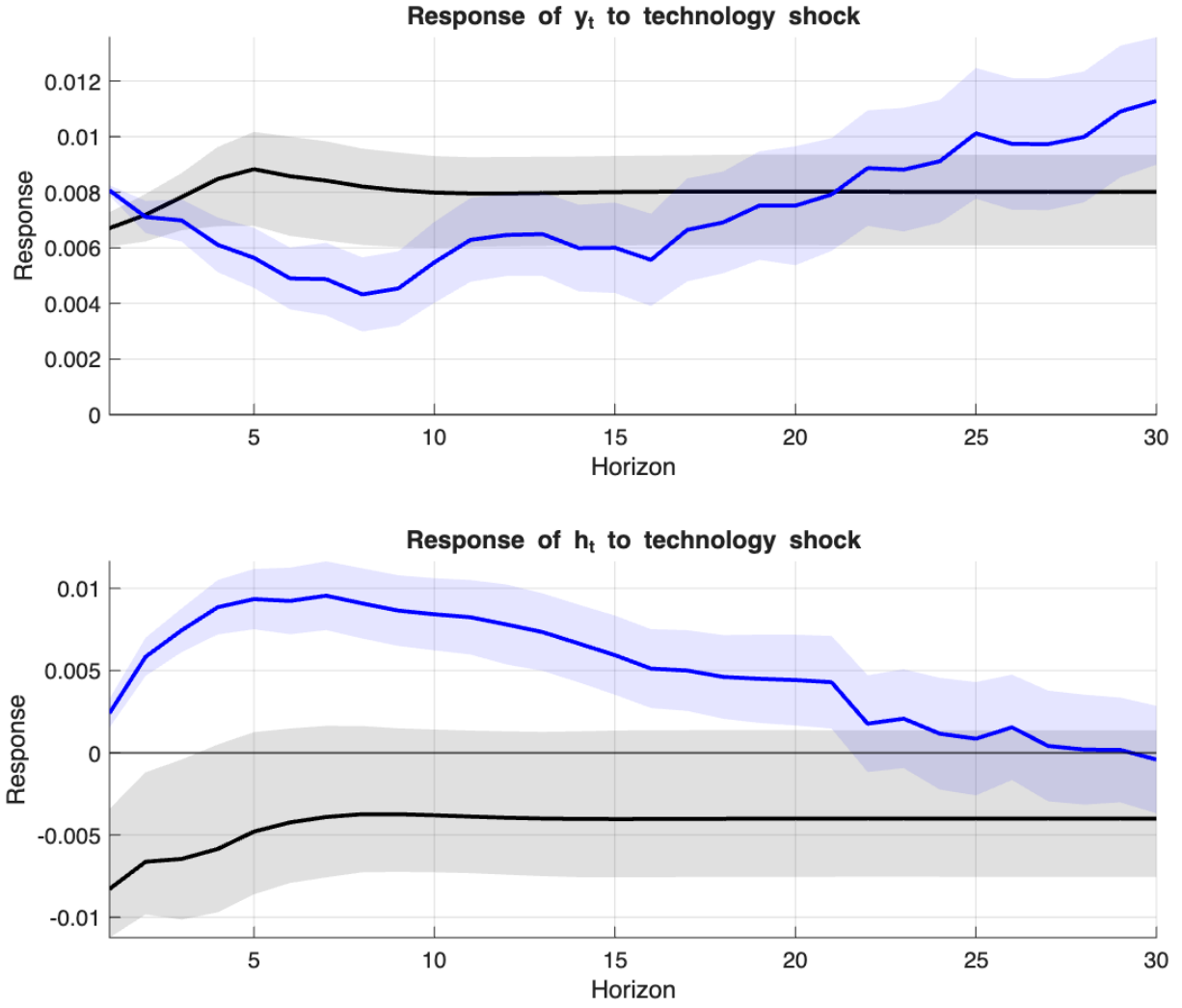


Figure 5: Cumulate Impulse Response and Linear Projection with $w_t = [\Delta y_t, \Delta h_t]'$
 Linear Projection are the Blue Lines with confidence interval in light-blue, HAC 68%
 Cumulative Impulse responses function from SVAR are in Dark with confidence interval in light-grey, Bootstrap 68%

IRF Second Specification

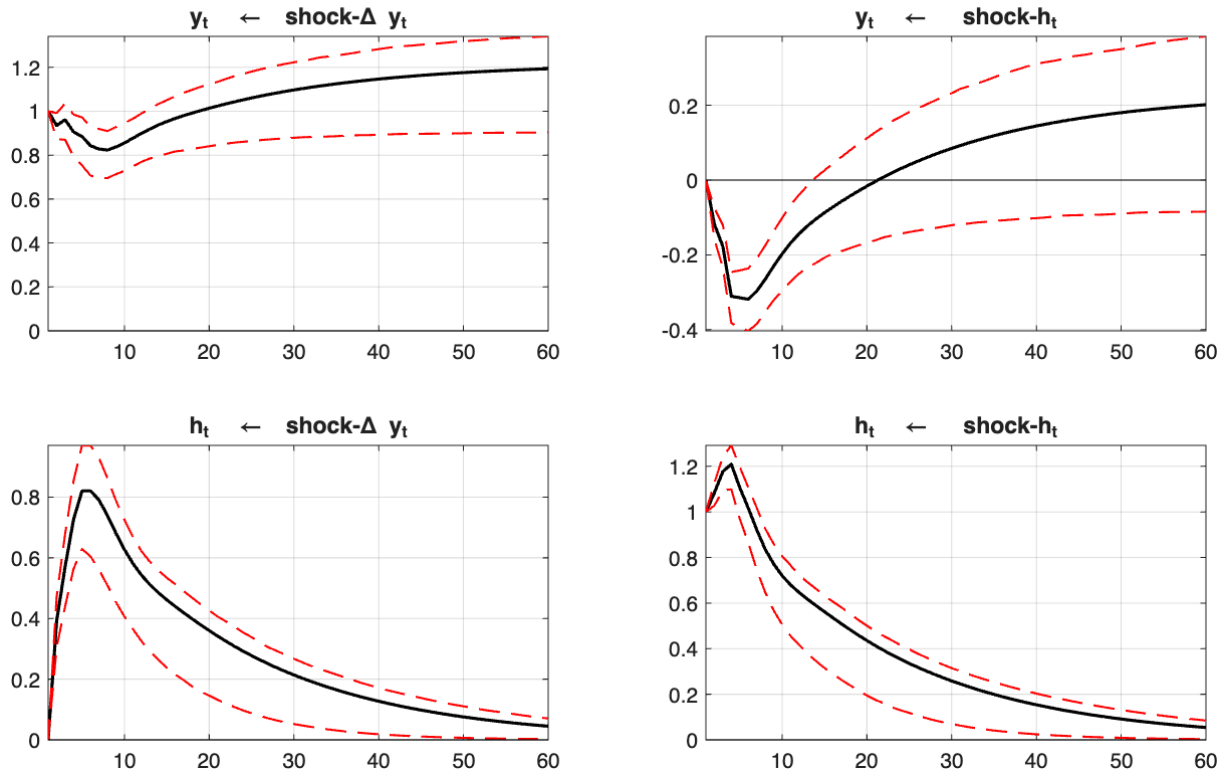


Figure 6: Cumulated Wold Impulse Responses from the VAR(4) with $w_t = [\Delta y_t, h_t]'$
Confidence Interval: Bootstrap 68%

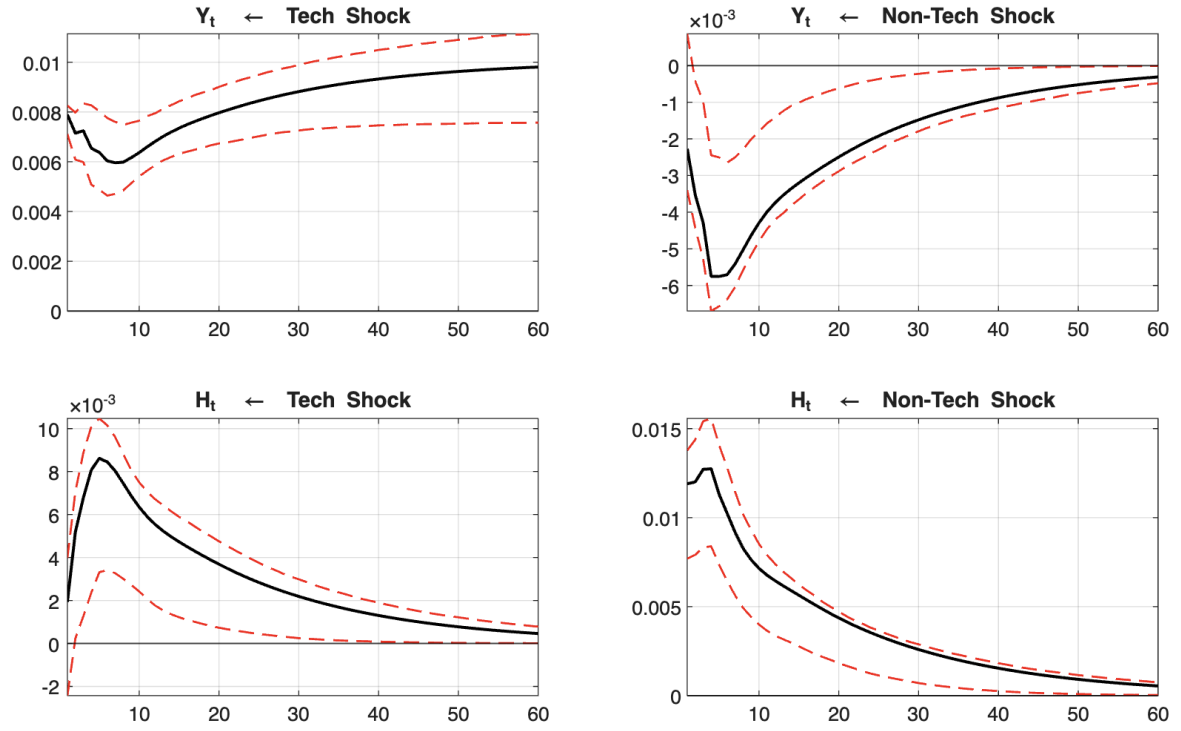


Figure 7: Impulse responses to the Identified Technology Shock with $w_t = [\Delta y_t, h_t]'$
Confidence Interval: Bootstrap 68%

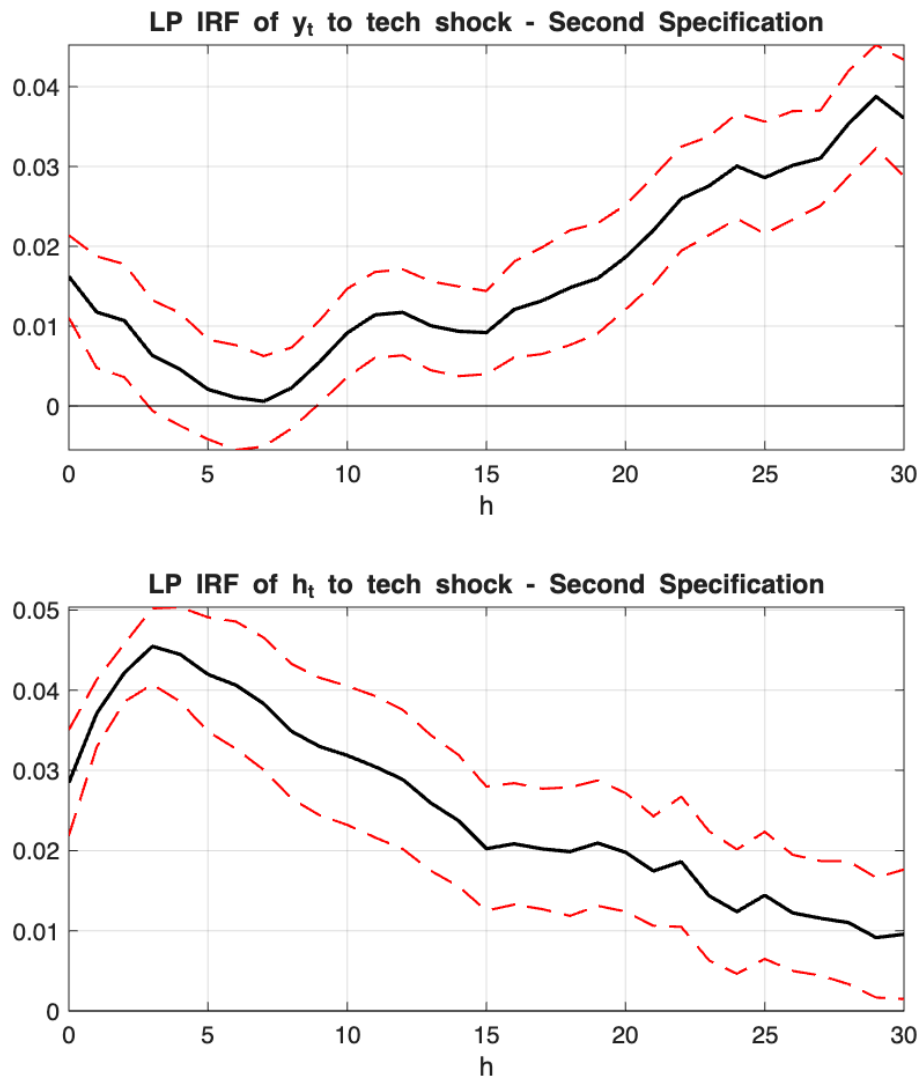


Figure 8: Linear Projection, for the second specification. Confidence Interval: HAC 68%

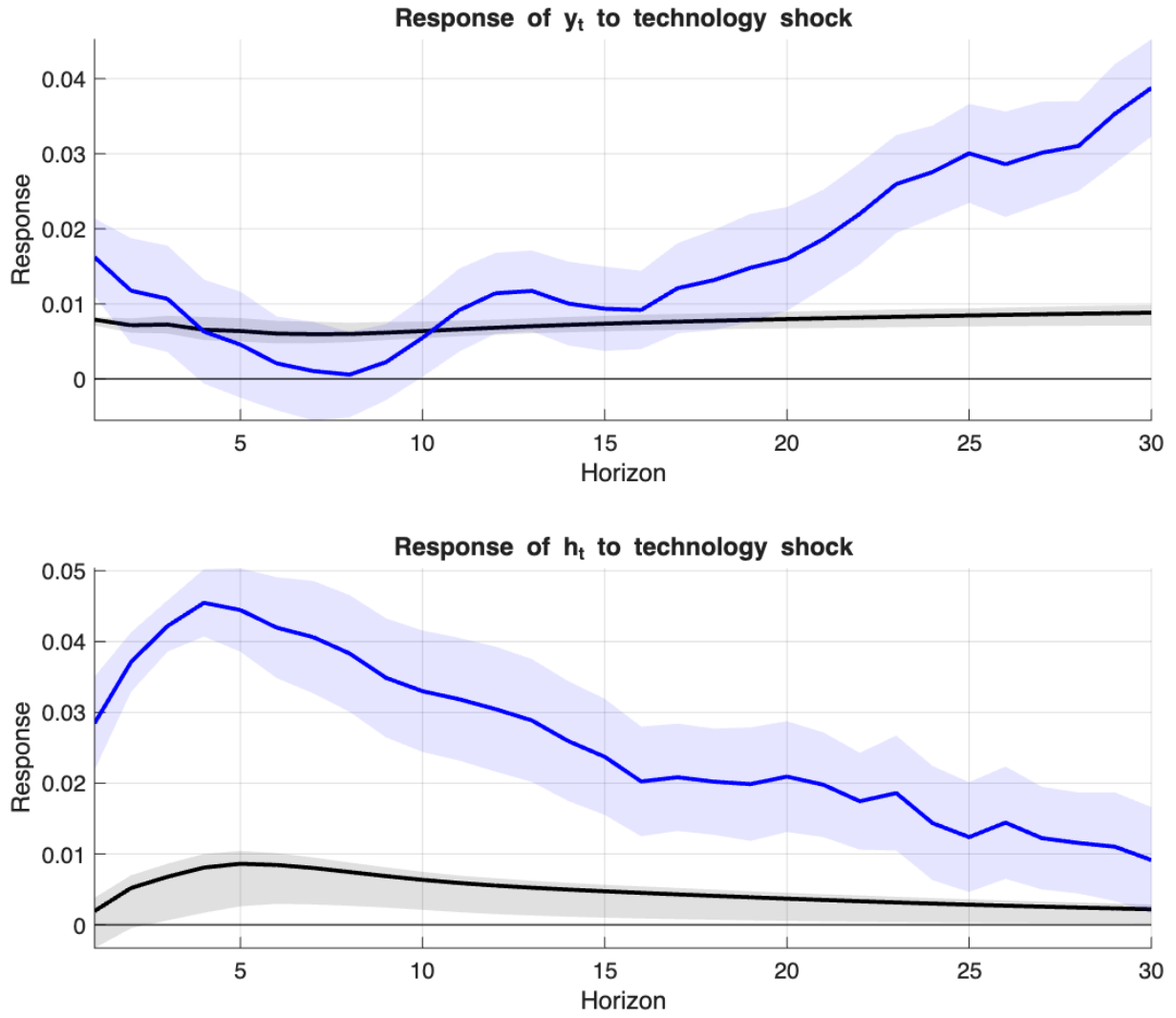


Figure 9: Cumulate Impulse Response and Linear Projection with $w_t = [\Delta y_t, h_t]'$
 Linear Projection are the Blue Lines with confidence interval in light-blue, HAC 68%
 Cumulative Impulse responses function from SVAR are in Dark with confidence interval in light-grey, Bootstrap 68%

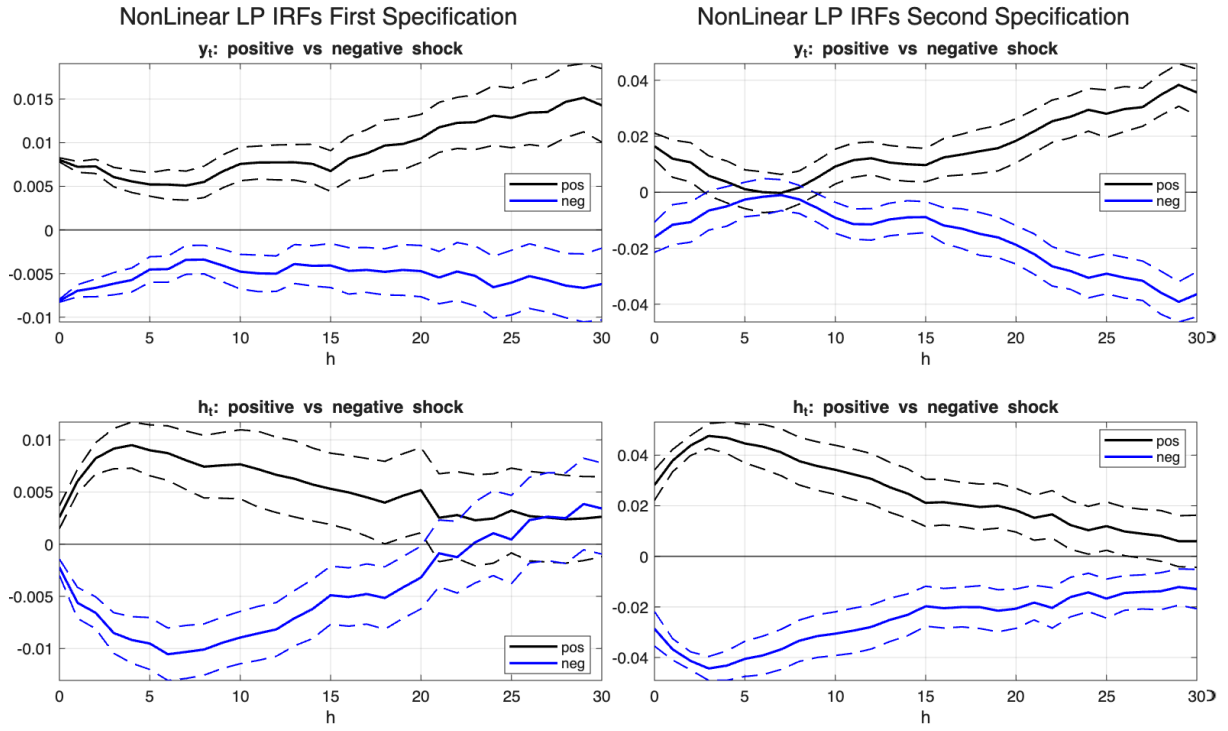


Figure 10: Sign-dependent LP IRFs for the two specifications -Confidence Interval: HAC 68%

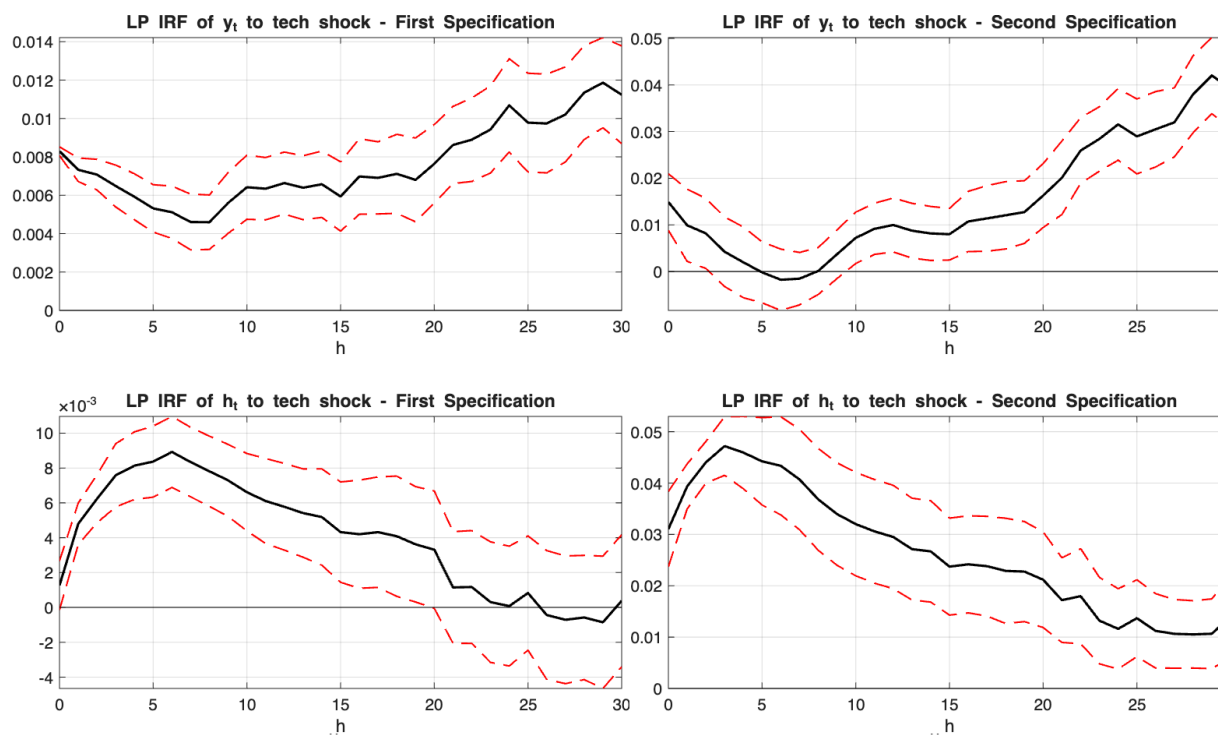


Figure 11: LP IRFs for the two specifications with 12 lags as control
Confidence Interval: HAC 68%

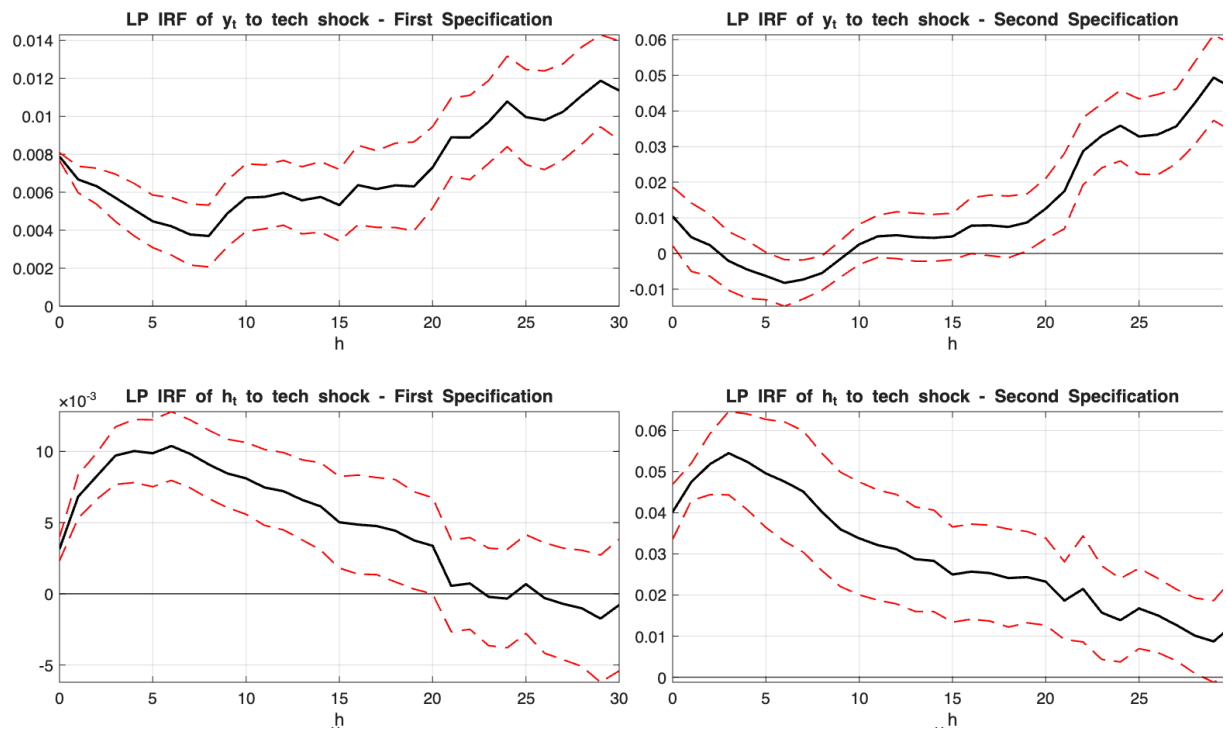


Figure 12: LP IRFs for the two specifications with 16 lags as control
Confidence Interval: HAC 68%

6.2 Tables

Table 1: Forecast Error Variance Decomposition (%) - Galí Specification

	Horizon (hor)			
	1	5	9	60
Δy_t	66.9319	62.5177	62.3093	62.2908
Δh_t	45.4770	43.0820	43.0854	43.0827

Table 2: Forecast Error Variance Decomposition (%) - CEV Specification

	Horizon (hor)			
	1	5	9	60
Δy_t	92.3527	87.0097	86.5745	85.9199
h_t	2.5765	22.6563	30.1351	34.6315

References

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